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WhatNext Vision Motors:

Shaping the Future of Mobility with Innovation and Excellence

Project Overview:

WhatsNext Vision Motors, a pioneering force in the automotive industry, is dedicated to transforming the mobility sector with innovative technology and solutions that prioritize customer needs. The company has embarked on an ambitious Salesforce project with the core objective of enhancing the customer experience and streamlining its operational processes.

Objectives:

- The project aims to enhance the overall customer experience while simultaneously streamlining the company's operational processes.
- The system intends to improve ordering convenience by automatically suggesting the nearest dealer location based on the customer's provided address.
- It seeks to prevent customer disappointment and improve order accuracy by restricting order placement to only those vehicles currently in stock.
- The project targets the automation of bulk order records to accurately update statuses to 'Pending' or 'Confirmed' based on real-time availability.
- A final objective is to reduce the administrative burden on staff, allowing them to focus on strategic tasks rather than manual processes.

Phase 1: Requirement Analysis & Planning

Business Requirements:

Customers: Require a seamless, low-effort purchasing journey where they can easily locate dealers near them and trust that the products they order are actually available.

Internal Staff: Need relief from manual data entry (assigning dealers) and manual status tracking (checking stock against orders), allowing them to focus on high-value tasks.

Scope and Objectives:

- Implementation of logic to calculate distance between Customer Address and Dealer Locations to auto-populate the "Nearest Dealer" field on the Order.
- Implementation of real-time stock checks to prevent creation of single orders for out-of-stock items.
- Development of a scheduled process to handle bulk order imports, checking inventory and setting status to 'Confirmed' or 'Pending' accordingly.

Phase 2: Salesforce Development - Backend & Configurations

Objects, Fields and Relationships

SETUP > OBJECT MANAGER

Vehicle Customer

Details

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

Field Label	Field Name	Data Type	Controlling Field	Indexed	
Address	Address__c	Text(50)			
Created By	CreatedById	Lookup(User)			
Email	Email__c	Email			
Last Modified By	LastModifiedById	Lookup(User)			
Owner	OwnerId	Lookup(User Group)		✓	
Phone	Phone__c	Phone			
Preferred Vehicle Type	Preferred_Vehicle_Type__c	Picklist			
Vehicle Customer Name	Name	Text(80)		✓	

SETUP > OBJECT MANAGER

Vehicle Dealer

Details

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

Field Label	Field Name	Data Type	Controlling Field	Indexed	
Created By	CreatedById	Lookup(User)			
Dealer Code	Dealer_Code__c	Auto Number			
Dealer Location	Dealer_Location__c	Text(50)			
Email	Email__c	Email			
Last Modified By	LastModifiedById	Lookup(User)			
Owner	OwnerId	Lookup(User Group)		✓	
Phone	Phone__c	Phone			
Vehicle Dealer Name	Name	Text(80)		✓	

SETUP > OBJECT MANAGER

Vehicle Order

Details

Fields & Relationships

9 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

Field Label	Field Name	Data Type	Controlling Field	Indexed	
Assigned Dealer	Assigned_Dealer__c	Lookup(Vehicle Dealer)		✓	
Created By	CreatedById	Lookup(User)			
Last Modified By	LastModifiedById	Lookup(User)			
Order Date	Order_Date__c	Date			
Owner	OwnerId	Lookup(User Group)		✓	
Status	Status__c	Picklist			
Vehicle	Vehicle__c	Lookup(Vehicle)		✓	
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)		✓	
Vehicle Order Number	Name	Auto Number		✓	

Setup > Object Manager

Vehicle Service Request

Details

Fields & Relationships

9 Items, Sorted by Field Label

Quick Find

NewDeleted FieldsField DependenciesSet History Tracking

Fields & Relationships

Field Label	Field Name	Data Type	Controlling Field	Indexed
Created By	CreatedById	Lookup(User)		
Issue Description	Issue_Description__c	Text(60)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User.Group)		✓
Service Date	Service_Date__c	Date		
Status	Status__c	Picklist		
Vehicle	Vehicle__c	Lookup(Vehicle)		✓
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)		✓
Vehicle Service Request Name	Name	Text(80)		✓

Setup > Object Manager

Vehicle Test Drive

Details

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find

NewDeleted FieldsField DependenciesSet History Tracking

Fields & Relationships

Field Label	Field Name	Data Type	Controlling Field	Indexed
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User.Group)		✓
Status	Status__c	Picklist		
Test Drive Date	Test_Drive_Date__c	Date		
Vehicle	Vehicle__c	Lookup(Vehicle)		✓
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)		✓
Vehicle Test Drive Name	Name	Text(80)		✓

Setup > Object Manager

Vehicle

Details

Fields & Relationships

9 Items, Sorted by Field Label

Quick Find

NewDeleted FieldsField DependenciesSet History Tracking

Fields & Relationships

Field Label	Field Name	Data Type	Controlling Field	Indexed
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User.Group)		✓
Price	Price__c	Currency(18, 0)		
Status	Status__c	Picklist		
Stock Quantity	Stock_Quantity__c	Number(18, 0)		
Vehicle Dealer	Vehicle_Dealer__c	Lookup(Vehicle Dealer)		✓
Vehicle Model	Vehicle_Model__c	Picklist		
Vehicle Name	Name	Text(80)		✓

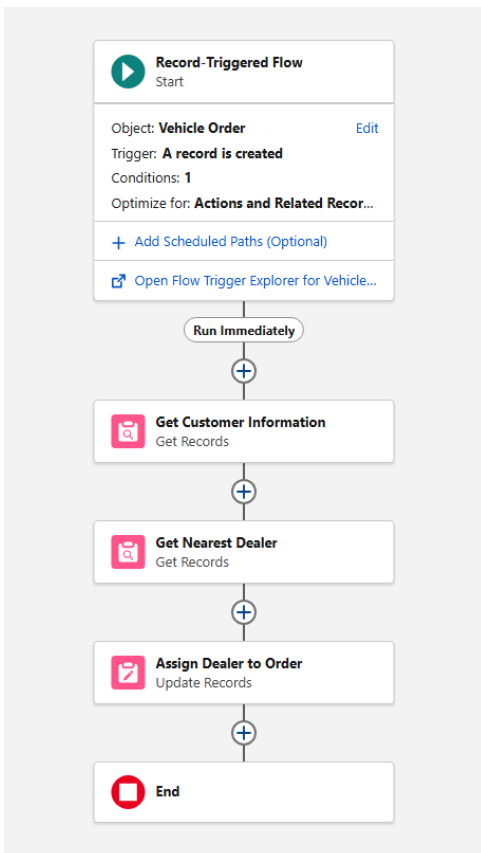
Lightning App

The screenshot displays the Lightning App interface. At the top, there's a navigation bar with the 'WhatNext Vision Motors' logo and a search bar. Below the navigation bar, the 'Vehicle Customers' section is active, showing a 'Recently Viewed' list. The list contains two items: 'Mark Mugen' and 'Ethan'. The interface includes various action buttons like 'New', 'Import', 'Change Owner', and 'Assign Label'. A search bar for the list is also present.

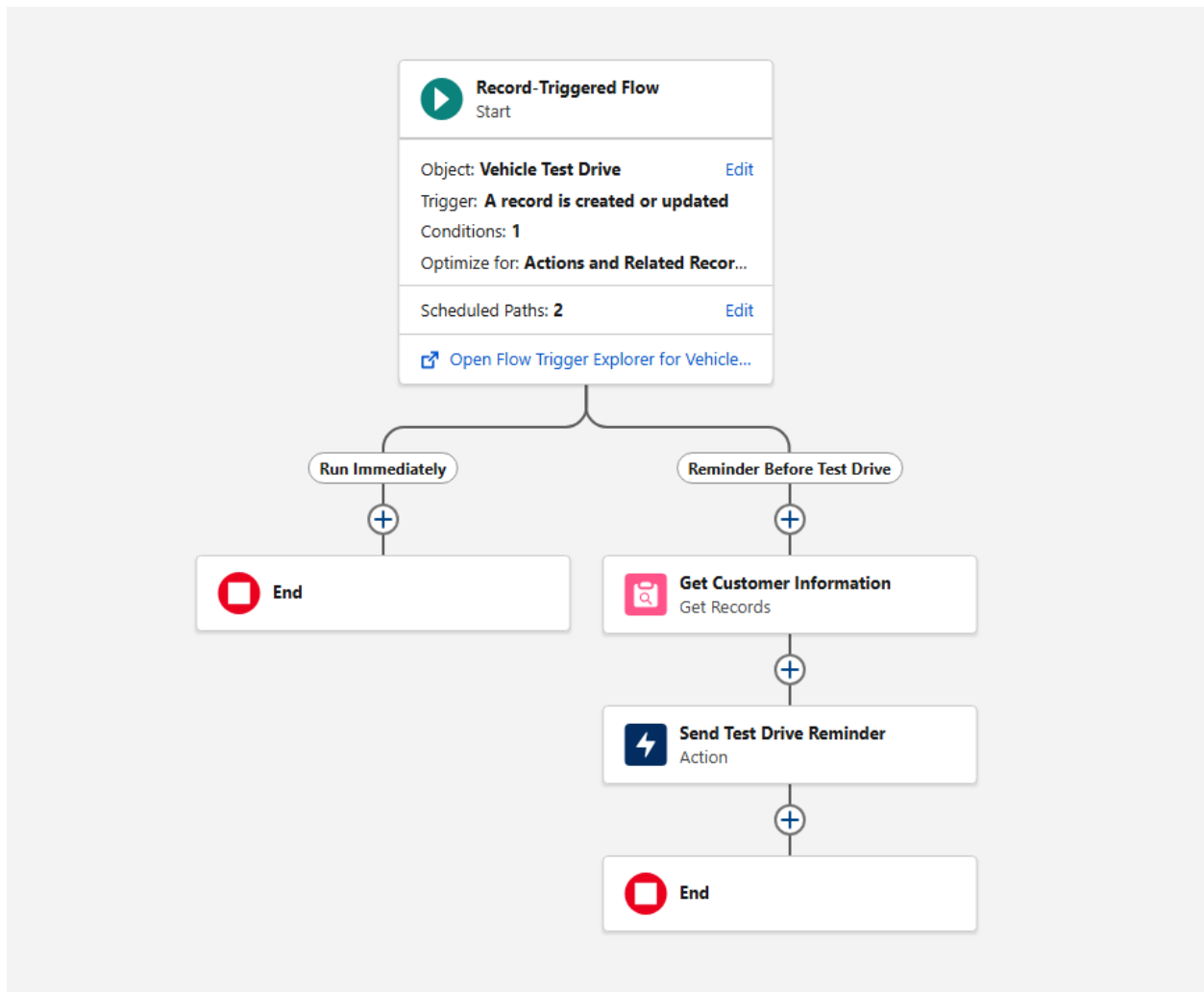
Vehicle Customer Name
1 ☐ Mark Mugen
2 ☐ Ethan

Flows

Auto Assign Dealer Flow

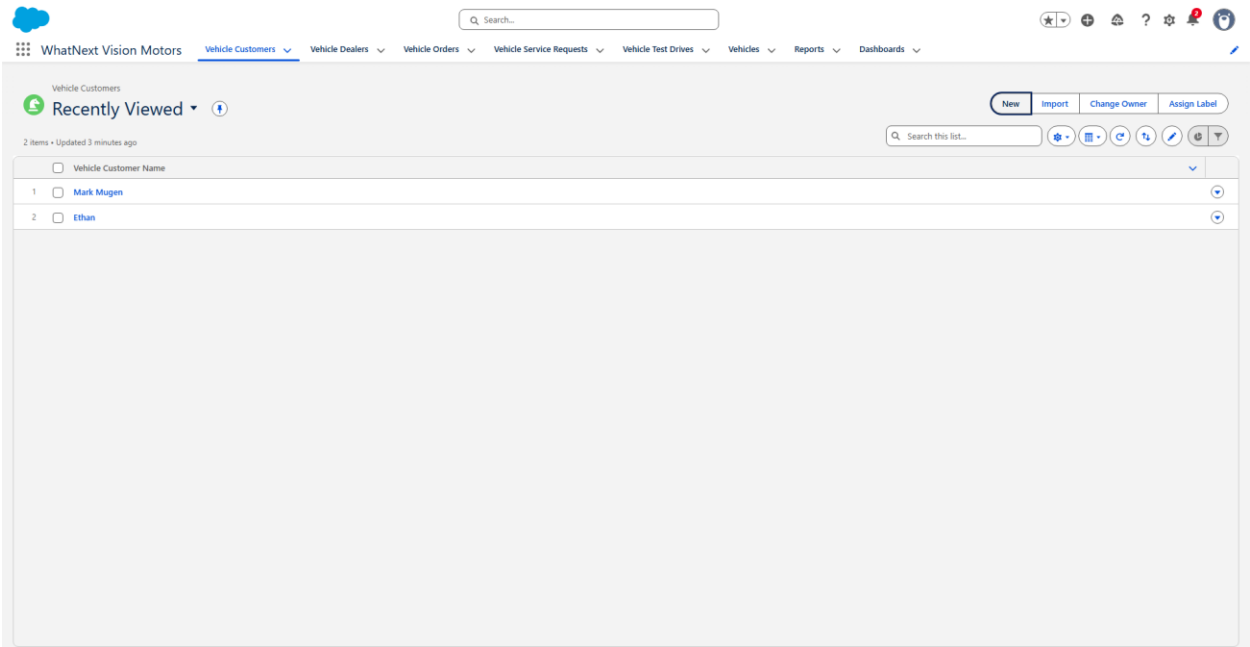


Test Drive Reminder Flow

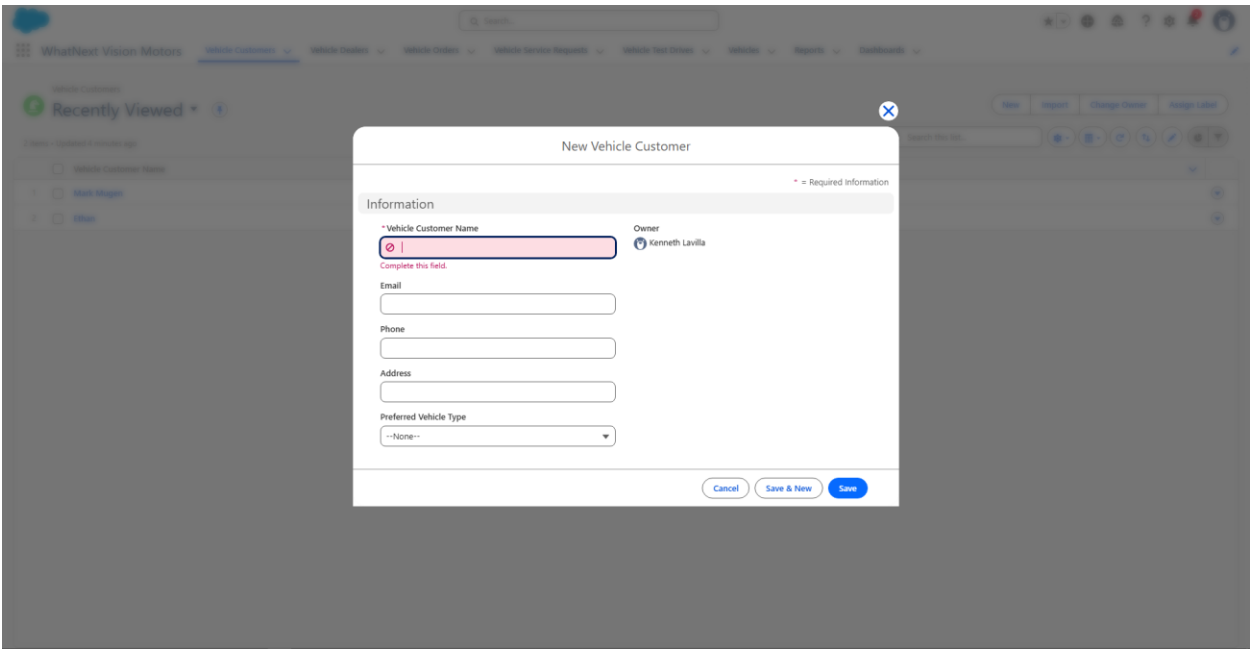


Phase 3: UI/UX Development & Customization

Lightning App



New Vehicle Customer Layout



New Vehicle Dealer Layout

The screenshot shows the 'New Vehicle Dealer' form within the 'WhatNext Vision Motors' application. The form is titled 'New Vehicle Dealer' and includes a close button (X) in the top right corner. A legend indicates that fields marked with an asterisk (*) are required information. The form contains the following fields:

- Vehicle Dealer Name**: A text input field with a red border and a red error message 'Complete this field.' below it.
- Owner**: A dropdown menu showing 'Kenneth Lavilla'.
- Dealer Location**: A text input field.
- Dealer Code**: A text input field.
- Phone**: A text input field.
- Email**: A text input field.

At the bottom of the form, there are three buttons: 'Cancel', 'Save & New', and 'Save'.

New Vehicle Order Layout

The screenshot shows the 'New Vehicle Order' form within the 'WhatNext Vision Motors' application. The form is titled 'New Vehicle Order' and includes a close button (X) in the top right corner. A legend indicates that fields marked with an asterisk (*) are required information. The form contains the following fields:

- Vehicle Order Number**: A text input field.
- Owner**: A dropdown menu showing 'Kenneth Lavilla'.
- Vehicle Customer**: A text input field with a search icon (Q) and a placeholder 'Search Vehicle Customers...'.
- Vehicle**: A text input field with a search icon (Q) and a placeholder 'Search Vehicles...'.
- Order Date**: A text input field with a calendar icon.
- Status**: A dropdown menu with 'None--' selected.
- Assigned Dealer**: A text input field with a search icon (Q) and a placeholder 'Search Vehicle Dealers...'.

At the bottom of the form, there are three buttons: 'Cancel', 'Save & New', and 'Save'.

New Vehicle Test Drive Layout

The screenshot shows the 'New Vehicle Service Request' form within the WhatNext Vision Motors application. The form is titled 'New Vehicle Service Request' and includes a close button (X) in the top right corner. A legend indicates that fields marked with an asterisk (*) are required information.

Information

- * Vehicle Service Request Name**: A text input field.
- Owner**: A dropdown menu showing 'Kenneth Lavilla'.
- Vehicle Customer**: A search input field with the placeholder 'Search Vehicle Customers...' and a magnifying glass icon.
- Vehicle**: A search input field with the placeholder 'Search Vehicles...' and a magnifying glass icon.
- Service Date**: A date input field with a calendar icon.
- Issue Description**: A text input field.
- Status**: A dropdown menu with '--None--' selected.

Buttons at the bottom: Cancel, Save & New, Save.

New Vehicle Layout

The screenshot shows the 'New Vehicle' form within the WhatNext Vision Motors application. The form is titled 'New Vehicle' and includes a close button (X) in the top right corner. A legend indicates that fields marked with an asterisk (*) are required information.

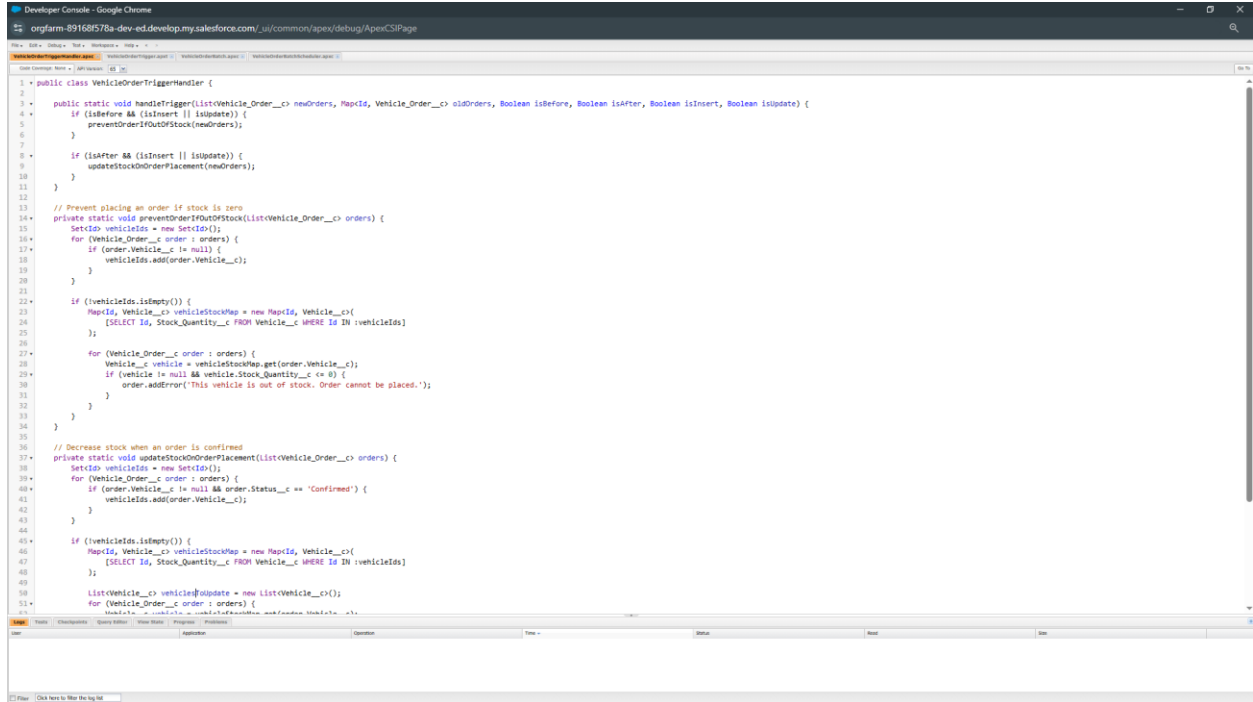
Information

- * Vehicle Name**: A text input field with a red error message 'Complete this field.' below it.
- Owner**: A dropdown menu showing 'Kenneth Lavilla'.
- Vehicle Model**: A dropdown menu with '--None--' selected.
- Stock Quantity**: A text input field.
- Price**: A text input field.
- Vehicle Dealer**: A search input field with the placeholder 'Search Vehicle Dealers...' and a magnifying glass icon.
- Status**: A dropdown menu with '--None--' selected.

Buttons at the bottom: Cancel, Save & New, Save.

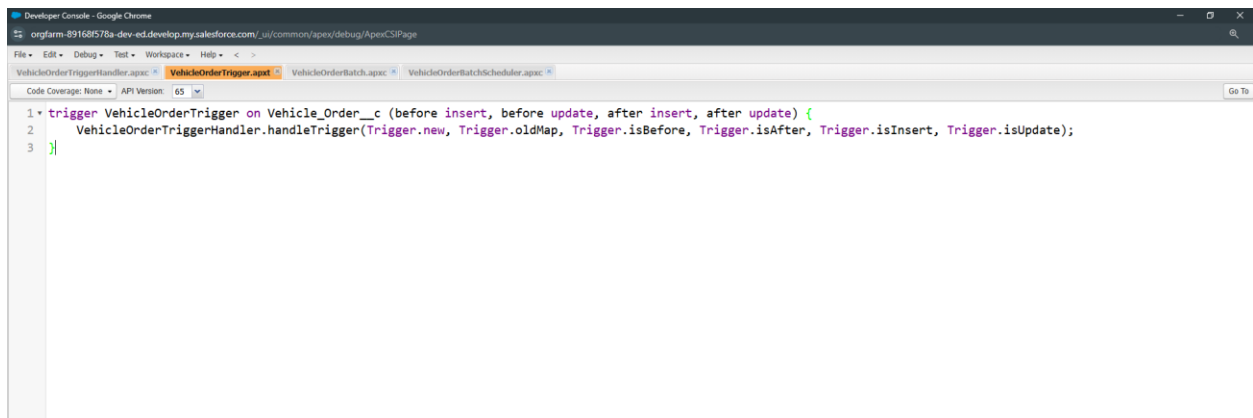
Phase 4: Data Migration, Testing & Security

Vehicle Order Trigger Handler



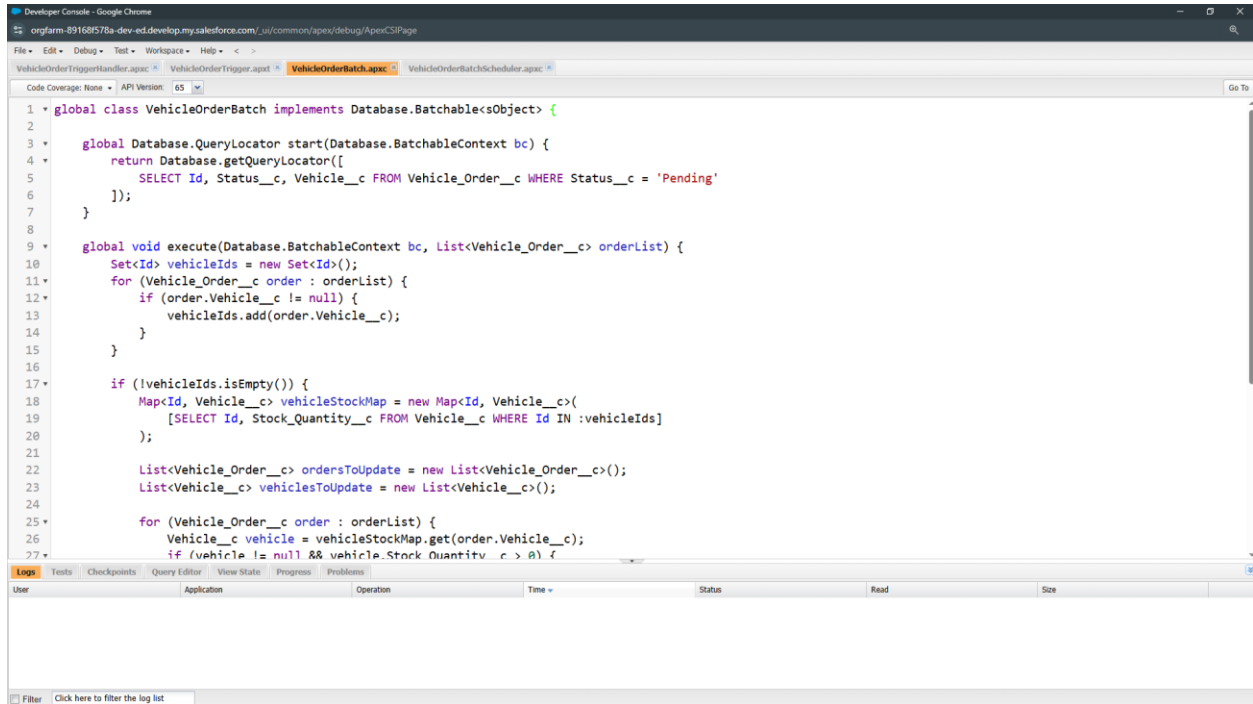
```
1 public class VehicleOrderTriggerHandler {
2
3     public static void handleTrigger(List<Vehicle_Order__c> newOrders, Map<Id, Vehicle_Order__c> oldOrders, Boolean isBefore, Boolean isAfter, Boolean isInsert, Boolean isUpdate) {
4         if (isBefore && (isInsert || isUpdate)) {
5             preventOrderIfOutOfStock(newOrders);
6         }
7
8         if (isAfter && (isInsert || isUpdate)) {
9             updateStockOnOrderPlacement(newOrders);
10        }
11    }
12
13    // Prevent placing an order if stock is zero
14    private static void preventOrderIfOutOfStock(List<Vehicle_Order__c> orders) {
15        Set<Id> vehicleIds = new Set<Id>();
16        for (Vehicle_Order__c order : orders) {
17            if (order.Vehicle__c != null) {
18                vehicleIds.add(order.Vehicle__c);
19            }
20        }
21
22        if (vehicleIds.isEmpty()) {
23            Map<Id, Vehicle__c> vehicleStockMap = new Map<Id, Vehicle__c>();
24            [SELECT Id, Stock_Quantity__c FROM Vehicle__c WHERE Id IN :vehicleIds];
25        }
26
27        for (Vehicle_Order__c order : orders) {
28            Vehicle__c vehicle = vehicleStockMap.get(order.Vehicle__c);
29            if (vehicle != null && vehicle.Stock_Quantity__c <= 0) {
30                order.addError('This vehicle is out of stock. Order cannot be placed.');
```

Vehicle Order Trigger



```
1 trigger VehicleOrderTrigger on Vehicle_Order__c (before insert, before update, after insert, after update) {
2     VehicleOrderTriggerHandler.handleTrigger(trigger.new, trigger.oldMap, trigger.isBefore, trigger.isAfter, trigger.isInsert, trigger.isUpdate);
3 }
```

Vehicle Order Batch

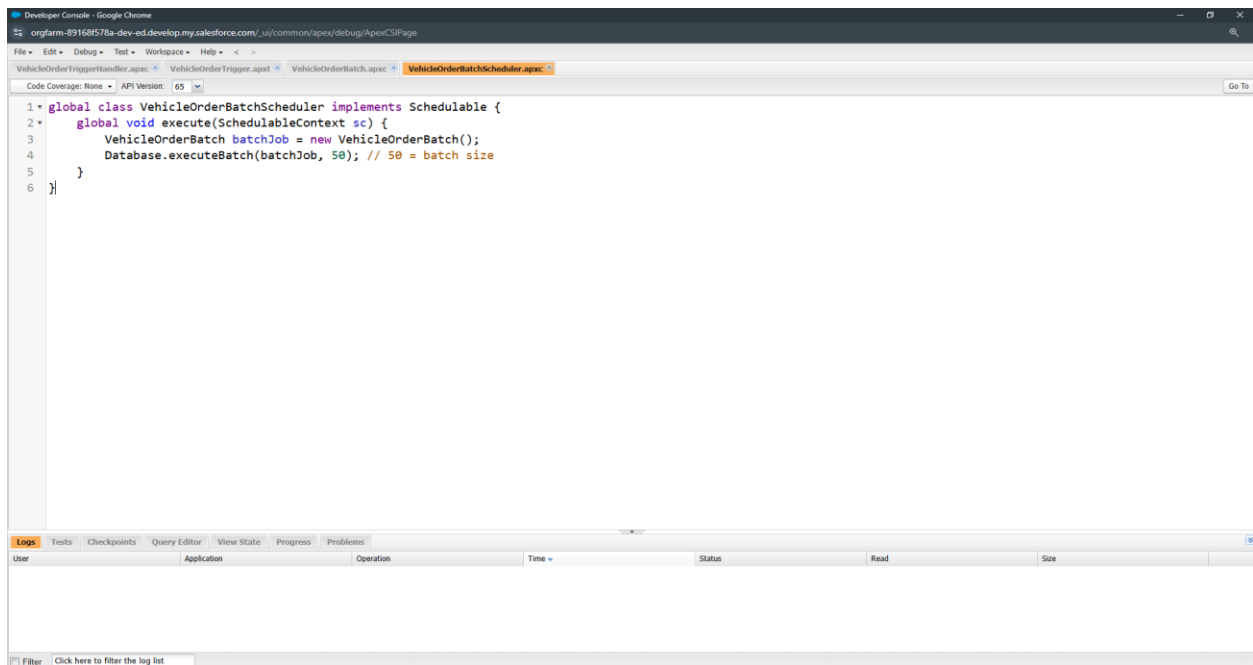


The screenshot shows the Salesforce Developer Console with the 'VehicleOrderBatch.apex' file open. The code defines a global class `VehicleOrderBatch` that implements `Database.Batchable<sObject>`. It includes a `start` method to get a query locator for pending orders, an `execute` method to process a list of orders by adding their IDs to a set, and logic to query stock quantities for those IDs and update the orders.

```
1 global class VehicleOrderBatch implements Database.Batchable<sObject> {
2
3     global Database.QueryLocator start(Database.BatchableContext bc) {
4         return Database.getQueryLocator([
5             SELECT Id, Status__c, Vehicle__c FROM Vehicle_Order__c WHERE Status__c = 'Pending'
6         ]);
7     }
8
9     global void execute(Database.BatchableContext bc, List<Vehicle_Order__c> orderList) {
10         Set<Id> vehicleIds = new Set<Id>();
11         for (Vehicle_Order__c order : orderList) {
12             if (order.Vehicle__c != null) {
13                 vehicleIds.add(order.Vehicle__c);
14             }
15         }
16
17         if (!vehicleIds.isEmpty()) {
18             Map<Id, Vehicle__c> vehicleStockMap = new Map<Id, Vehicle__c>{
19                 [SELECT Id, Stock_Quantity__c FROM Vehicle__c WHERE Id IN :vehicleIds]
20             };
21
22             List<Vehicle_Order__c> ordersToUpdate = new List<Vehicle_Order__c>();
23             List<Vehicle__c> vehiclesToUpdate = new List<Vehicle__c>();
24
25             for (Vehicle_Order__c order : orderList) {
26                 Vehicle__c vehicle = vehicleStockMap.get(order.Vehicle__c);
27                 if (vehicle != null && vehicle.Stock_Quantity__c > 0) {
```

Below the code editor, there is a 'Logs' tab and a table with columns: User, Application, Operation, Time, Status, Read, Size. A filter button is also present.

Vehicle Order Batch Scheduler



The screenshot shows the Salesforce Developer Console with the 'VehicleOrderBatchScheduler.apex' file open. The code defines a global class `VehicleOrderBatchScheduler` that implements `Schedulable`. It includes an `execute` method that creates a `VehicleOrderBatch` instance and calls `Database.executeBatch` with a batch size of 50.

```
1 global class VehicleOrderBatchScheduler implements Schedulable {
2     global void execute(SchedulableContext sc) {
3         VehicleOrderBatch batchJob = new VehicleOrderBatch();
4         Database.executeBatch(batchJob, 50); // 50 = batch size
5     }
6 }
```

Below the code editor, there is a 'Logs' tab and a table with columns: User, Application, Operation, Time, Status, Read, Size. A filter button is also present.

Auto Assign Dealer Flow Test

WhatNext Vision Motors

Vehicle Customers

Vehicle Dealers

Vehicle Orders

Vehicle Service Requests

Vehicle Test Drives

Vehicles

Reports

Dashboards

Vehicle Order

O-0005

New Contact

Edit

New Opportunity

Related

Details

Vehicle Order Number

O-0005

Owner

Kenneth Lavilla

Vehicle Customer

Mark Muyen

Vehicle

Honda

Order Date

11/26/2025

Status

Pending

Assigned

QWert

Created

Kenneth Lavilla, 11/25/2025, 5:28 AM

Last Modified By

Kenneth Lavilla, 11/25/2025, 5:28 AM

Vehicle Order Trigger Handler Test

The initial stock is 49.

Vehicle

Honda

Related

Details

Vehicle Name

Honda

Owner

Kenneth Lavilla

Vehicle Model

EV

Stock Quantity

49

Price

\$50,000

Vehicle Dealer

Honda

Status

Available

Created By

Kenneth Lavilla, 11/25/2025, 5:28 AM

Last Modified By

Kenneth Lavilla, 11/25/2025, 5:36 AM

Creation of new vehicle order

Vehicle Dealers Vehicle Orders Vehicle Service Requests Vehicle Test Drives Vehicles Reports Dashboards

New Vehicle Order

* = Required Information

Information

Vehicle Order Number

Owner
Kenneth Lavilla

Vehicle Customer
Mark Mugen

Vehicle
Honda

Order Date
11/26/2025

Status
Confirmed

Assigned Dealer
Search Vehicle Dealers...

Cancel Save & New Save

Vehicle Order
O-0007

Related Details

Vehicle Order Number O-0007	Owner Kenneth Lavilla
Vehicle Customer Mark Mugen	
Vehicle Honda	
Order Date 11/26/2025	
Status Confirmed	
Assigned Dealer	
Created By Kenneth Lavilla, 11/25/2025, 5:37 AM	Last Modified By Kenneth Lavilla, 11/25/2025, 5:37 AM

The stock became 48 after creating a new vehicle order.

The screenshot shows a 'Vehicle Details' page for a Honda EV. The page has a header with the 'Vehicle' icon and 'Honda' text. Below the header, there are two tabs: 'Related' and 'Details', with 'Details' being the active tab. The details are organized into two columns. The left column contains fields for 'Vehicle Name' (Honda), 'Vehicle Model' (EV), 'Stock Quantity' (48), 'Price' (\$50,000), 'Vehicle Dealer' (Honda), 'Status' (Available), and 'Created By' (Kenneth Lavilla, 11/25/2025, 5:28 AM). The right column contains 'Owner' (Kenneth Lavilla) and 'Last Modified By' (Kenneth Lavilla, 11/25/2025, 5:37 AM). Each field has an edit icon (pencil) next to it.

Field	Value
Vehicle Name	Honda
Vehicle Model	EV
Stock Quantity	48
Price	\$50,000
Vehicle Dealer	Honda
Status	Available
Created By	Kenneth Lavilla, 11/25/2025, 5:28 AM
Owner	Kenneth Lavilla
Last Modified By	Kenneth Lavilla, 11/25/2025, 5:37 AM

Error Test if the stock 0 and tried to create a new vehicle order.

The screenshot shows a 'New Vehicle Order' form. The form has a title bar with a close button (X). Below the title bar, there is a section for 'Information' with a sub-header 'Information'. The form contains several fields: 'Vehicle Order Number', 'Vehicle Customer' (Mark Mugen), 'Vehicle' (Honda), 'Order Date' (11/26/2025), 'Status' (Confirmed), and 'Assigned Dealer' (Search Vehicle Dealers...). The 'Owner' field is set to 'Kenneth Lavilla'. A red error message box is displayed over the form, stating 'We hit a snag. Review the errors on this page. This vehicle is out of stock. Order cannot be placed.' The form has buttons for 'Cancel', 'Save & New', and 'Save'.

New Vehicle Order

* = Required Information

Information

Vehicle Order Number

Vehicle Customer: Mark Mugen

Vehicle: Honda

Order Date: 11/26/2025

Status: Confirmed

Assigned Dealer: Search Vehicle Dealers...

Owner: Kenneth Lavilla

We hit a snag.

Review the errors on this page.

- This vehicle is out of stock. Order cannot be placed.

Cancel Save & New Save

THANK YOU!